SpikeDistill-RUL Project Report

PS 1 · Turbofan remaining useful life estimation

Abstract

We study whether an attention BiLSTM can transfer useful degradation patterns to a compact recurrent spiking student. Experiments cover NASA C-MAPSS FD001–FD004, 12 baseline entries and four student variants. SpikeDistill-RUL improves on the fixed LIF baseline on all four subsets, with about 89% fewer parameters than its teacher. Conventional models lead mean raw test RMSE. Additional experiments examine ensembles, student width and missing sensors. The FD001 ONNX export passed CPU prediction checks. Physical Jetson measurements remain pending.

1. Background and related work

RUL is the number of operating cycles left before failure. Predicting it from incomplete sensor histories can support maintenance scheduling. An optimistic estimate can delay action, which motivates reporting the asymmetric NASA score alongside standard regression errors.

NASA C-MAPSS [1] provides simulated engine degradation histories. LSTM-based RUL estimation [2] and the recurrent architectures in [3–5,16] motivate our sequence baselines. Attention-based RUL [6] motivates the teacher, while convolutional RUL features [17] provide context for CNN-LSTM.

Surrogate gradients [7,8] enable training with hard forward spikes. DIET-SNN [9] motivates learnable leak and threshold, and ANN-to-SNN conversion [10] provides a rate-based control. Output distillation [11] and intermediate hints [12] motivate matching teacher predictions and hidden states. We also match temporal attention. IG [13], SHAP [14] and TimeSHAP [15] provide the background for the explanation methods.

Identified research gaps

Value of each distillation signal. Matching the teacher’s final prediction leaves open whether its hidden states and attention provide useful additional guidance. We tested each signal by removing its loss while keeping the other training settings fixed.

Student size versus prediction quality. The original 48-neuron student gave one point on the size–accuracy curve. We tested four widths and compared inference parameters, RMSE and the NASA score.

Combining independently trained students. Reporting variation across seeds leaves open whether averaging their predictions helps. We compared equal and validation-weighted ensembles with the validation-selected single student.

Sensitivity to missing sensors. Clean test windows do not capture channel outages. We simulated one missing sensor at a time and compared the teacher and student on the same affected windows.

The assignment specified the core teacher-student design. Our contribution is its controlled implementation and evaluation, separate loss ablations, and the additional studies of ensemble stability, student capacity and sensor outages.

2. Dataset, preprocessing and baselines

Subset	Train engines	Test engines	Training windows	Features
FD001	100	100	14,459	18
FD002	260	259	37,051	24
FD003	100	100	17,754	19
FD004	249	248	42,960	24

Table 1. Official engine counts before the validation holdout. FD004 has 248 aligned test engines in the supplied files.

Split seed 42 holds out approximately 20% of training engines. Sensor selection and scaling use training engines only. FD001/FD003 use global sensor standardization, while FD002/FD004 use six training-fitted operating regimes.

Inputs contain 30 cycles with stride one for training. Short histories repeat their first row on the left and use a validity mask. Validation uses fixed incomplete prefixes. Test scoring uses one final window per official engine.

Fit targets are min(RUL,125)/125. Checkpoints minimize capped validation RMSE. Test predictions are restored to cycles without clipping and scored against raw labels. The split is shared across all model seeds.

Baseline	Configuration
Mean	Training-label mean
DeepNN / WideNN	Flattened window; 128–96–64–32 ReLU layers / one 256-unit ReLU layer
RNN / LSTM / GRU	One 48-unit recurrent layer; final-state readout
BiLSTM	48 units in each direction
CNN-LSTM	48-channel Conv1D, kernel 3; 48-unit LSTM
Attention BiLSTM	BiLSTM with masked temporal attention
Small LSTM with KD	16-unit LSTM with label and teacher-output MSE
Fixed LIF / ANN-to-SNN	48 fixed LIF neurons / calibrated DeepNN conversion with 32 steps

All neural fits use Adam at 0.001, batch size 128, 30 epochs, gradient clipping at 1.0 and seeds 42–44.

3. SpikeDistill-RUL methodology

The validation-selected attention BiLSTM is frozen. It provides a scalar prediction, 96-dimensional hidden states and temporal attention. The student has 48 recurrent LIF neurons. Sensor values supply input current, and preceding spikes provide recurrent feedback.

At a valid cycle, $\tilde{v}_t = \lambda \odot v_{t-1} + W_i x_t + b_i + W_r s_{t-1}$, $s_t = 1[\tilde{v}_t \geq \theta]$, and $v_t = \tilde{v}_t - \theta \odot s_t$. Leak and threshold are learned per neuron through sigmoid and softplus transforms, initialized at 0.9 and 1.0. Backpropagation uses the surrogate $(1 + 10|\tilde{v}_t - \theta|)^{-2}$ and detaches the spike in the reset. Padding keeps the state unchanged.

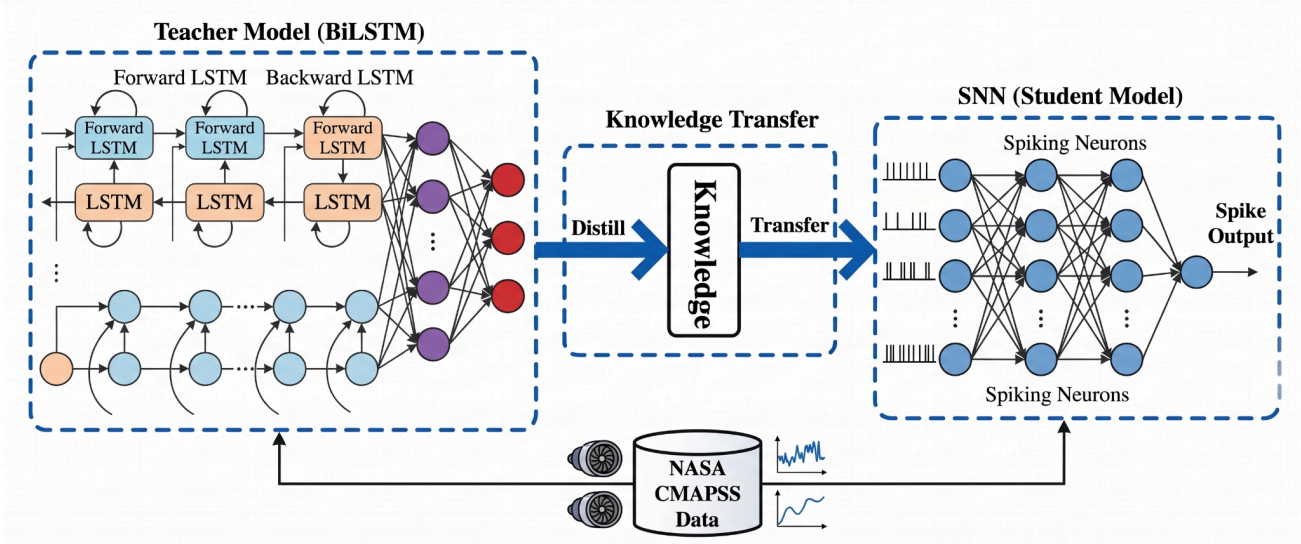
Attention computed from membrane states weights the spikes before a linear RUL readout. A width-to-96 adapter matches student membranes to teacher states during training. The teacher and adapter are excluded from student inference.

Student variant	Label	Output	Hidden	Attention	Firing rate
Learnable SNN	1.0	0	0	0	0
Output KD	0.5	0.5	0	0	0
Temporal KD	0.5	0.5	0.1	0.1	0
SpikeDistill-RUL	0.5	0.5	0.1	0.1	0.01

Table 2. Loss coefficients. The first four terms are MSE losses. Firing rate is averaged over valid neuron-cycle slots.

The full loss is 0.5 Llabel + 0.5 Loutput + 0.1 Lhidden + 0.1 Lattention + 0.01 ρ. Hidden and attention losses use valid positions only. The student uses 3,409–3,697 inference parameters, compared with 30,914–33,218 for the teacher.

Training procedure



4. Official test results

MAE, MSE, RMSE, R^2 and RAE are calculated in restored cycle units. For error e = prediction − target, the NASA score sums exp(−e/13)−1 when e < 0 and exp(e/10)−1 otherwise. Overestimation receives a steeper penalty. Compare summed scores within the same subset.

Model	FD001	FD002	FD003	FD004
Mean predictor	41.98 ± 0.00	53.78 ± 0.00	43.76 ± 0.00	54.56 ± 0.00
DeepNN	16.66 ± 0.46	28.11 ± 0.55	15.03 ± 0.58	27.84 ± 0.41
WideNN	16.98 ± 0.53	29.52 ± 1.38	16.77 ± 0.47	28.41 ± 1.28
RNN	16.21 ± 0.53	29.15 ± 0.70	16.79 ± 1.41	27.66 ± 0.52
LSTM	17.07 ± 0.81	28.90 ± 0.87	17.86 ± 1.21	26.70 ± 0.37
GRU	16.11 ± 0.50	29.23 ± 1.06	15.81 ± 0.57	26.11 ± 0.40
BiLSTM	15.75 ± 0.52	28.29 ± 0.36	15.32 ± 0.09	27.81 ± 0.61
CNN-LSTM	17.28 ± 0.73	30.59 ± 1.22	16.88 ± 0.60	27.67 ± 0.81
Attention BiLSTM	15.72 ± 0.00	29.05 ± 0.64	14.70 ± 0.49	27.10 ± 0.54
Small LSTM + KD	15.83 ± 0.44	29.39 ± 0.56	15.53 ± 1.48	26.75 ± 0.31
Fixed LIF SNN	19.66 ± 2.18	30.23 ± 0.70	21.69 ± 0.08	30.84 ± 0.93
ANN-to-SNN	44.18 ± 5.59	57.58 ± 8.30	31.65 ± 3.23	53.69 ± 6.30
Learnable LIF	18.29 ± 0.71	29.44 ± 0.58	20.01 ± 1.71	29.27 ± 0.99
Spiking output KD	17.37 ± 0.36	29.06 ± 0.74	20.32 ± 1.82	29.21 ± 0.42
Temporal spiking KD	17.07 ± 0.98	29.65 ± 0.03	19.07 ± 1.37	28.47 ± 0.87
SpikeDistill-RUL	17.67 ± 0.11	29.00 ± 0.06	20.18 ± 0.51	28.33 ± 0.79

Table 3. Raw test RMSE in cycles, mean ± sample SD over seeds 42–44.

Subset	MAE	MSE	R^2	RAE (%)	NASA score
FD001	13.65	312.14	0.819	37.11	588.42
FD002	19.49	841.16	0.709	43.04	15,255.02
FD003	15.43	407.57	0.762	43.11	1,439.58
FD004	21.30	802.77	0.730	45.83	6,090.46

Table 4. Additional raw test scores for SpikeDistill-RUL, averaged across three seeds.

Attention BiLSTM leads RMSE on FD001/FD003, DeepNN on FD002 and GRU on FD004. SpikeDistill-RUL improves on fixed LIF across all subsets. Its main demonstrated advantage over the teacher is model size.

5. Distillation ablations

A separate matched CPU experiment retrained the full student and two variants, each removing one loss. Teacher, architecture, data, budget, attention pooling and spike penalty were fixed within each subset and seed.

Subset	Full control	Without hidden matching	Without attention matching
FD001	17.07 ± 1.13	17.46 ± 0.85	17.26 ± 1.18
FD002	29.56 ± 0.09	29.91 ± 0.88	30.18 ± 0.37
FD003	19.88 ± 0.46	21.26 ± 0.87	20.48 ± 0.97
FD004	28.81 ± 0.20	28.84 ± 1.03	29.09 ± 0.57

Table 5. Raw test RMSE, mean ± sample SD over three seeds. These full controls are separate from the original Colab runs.

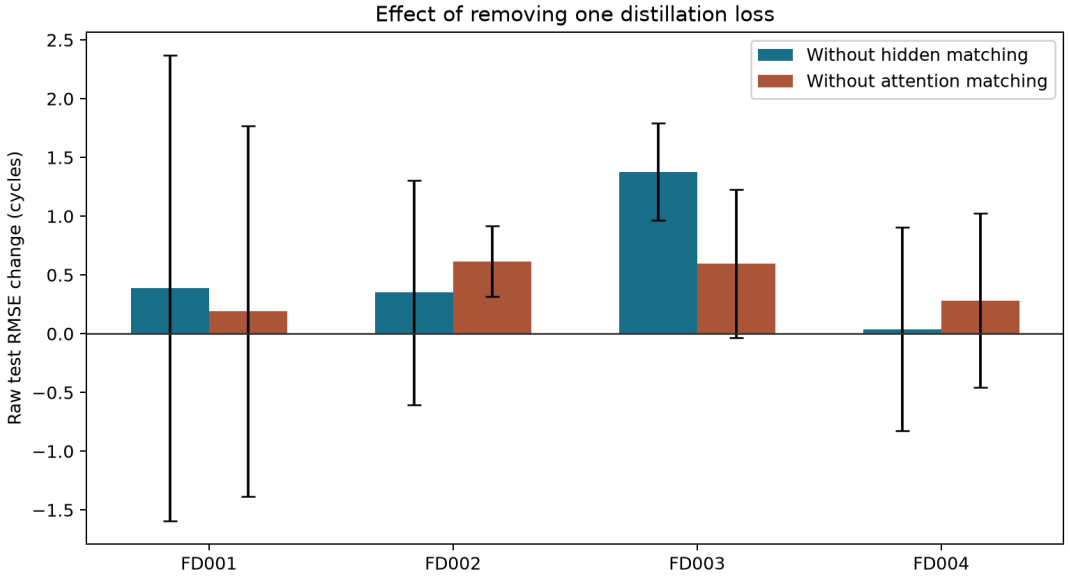


Figure 1. RMSE increase after removing a loss. Error bars are sample SD of paired seed differences.

Removing hidden matching increased FD003 RMSE by 1.38 cycles, the strongest observed effect. The FD004 increase was only 0.04 cycles, and its NASA score favored attention removal. These results show subset dependence. The original sparsity comparison was also mixed, improving RMSE on FD002/FD004 and increasing it on FD001/FD003.

6. Additional research — ensembles and capacity

Our first extension asks whether combining trained seeds improves a deployable predictor. Each subset uses the three frozen full controls from the ablation study. Equal averaging is compared with inverse-validation-MSE weighting and the validation-selected best single member. Member selection and weights were fixed before test scoring.

Subset	Validation-selected single	Equal ensemble	Validation-weighted ensemble
FD001	15.970	16.014	15.899
FD002	29.646	28.879	28.886
FD003	20.365	19.033	19.060
FD004	28.652	28.130	28.128

Table 6. Raw test RMSE. One ensemble per subset uses three members.

Validation weighting improved RMSE over the selected single model on all four subsets. Equal averaging slightly worsened FD001 RMSE. Both methods reduced NASA scores on every subset. FD001 validation weighting reduced the score from 689.19 to 484.14, but requires 10,227 inference parameters instead of 3,409. Device cost remains to be measured.

The second extension asks whether 48 neurons is a useful capacity choice. Students of width 16, 32, 48 and 64 were trained on FD001 at seed 42 with the same teacher, full loss and 30-epoch budget.

Width	Inference parameters	Raw test RMSE	NASA score
16	625	17.750	641.24
32	1,761	17.302	527.11
48	3,409	15.970	689.19
64	5,569	18.322	735.22

Table 7. FD001 capacity results at seed 42. The 48-neuron student had the lowest RMSE. The 32-neuron student had the lowest NASA score with about 48% fewer parameters. A width of 64 increased RMSE. More seeds and subsets are needed before selecting a generally preferred width.

7. Sensor outages and explainability

The third extension tests missing sensors. Using the original seed-42 teacher and student, we replaced one sensor's entire standardized trace with zero, equivalent to imputing its training mean. Other sensors, settings and masks were preserved. Both models received identical affected windows.

Subset	Teacher clean RMSE	Largest teacher increase	Student clean RMSE	Largest student increase
FD001	15.729	+1.945 (S02)	17.548	+4.138 (S14)
FD002	29.698	+2.666 (S14)	28.962	+3.932 (S02)
FD003	14.166	+5.696 (S12)	20.402	+7.258 (S12)
FD004	27.695	+3.201 (S11)	28.492	+5.975 (S11)

Table 8. Raw RMSE and largest outage-related increase. Each model's largest increase may involve a different sensor.

The student's largest RMSE increase exceeded the teacher's on every subset. Sensor 12 was most harmful for both on FD003 and sensor 11 on FD004. This is evidence of sensitivity under a specified imputation scenario at one seed.

We also used temporal Shapley values, surrogate Integrated Gradients, attention maps, spike contributions and sensor permutation. Shapley sampling uses five-cycle sensor blocks. The surrogate IG maps have large completeness residuals and are interpreted as approximate sensitivity.

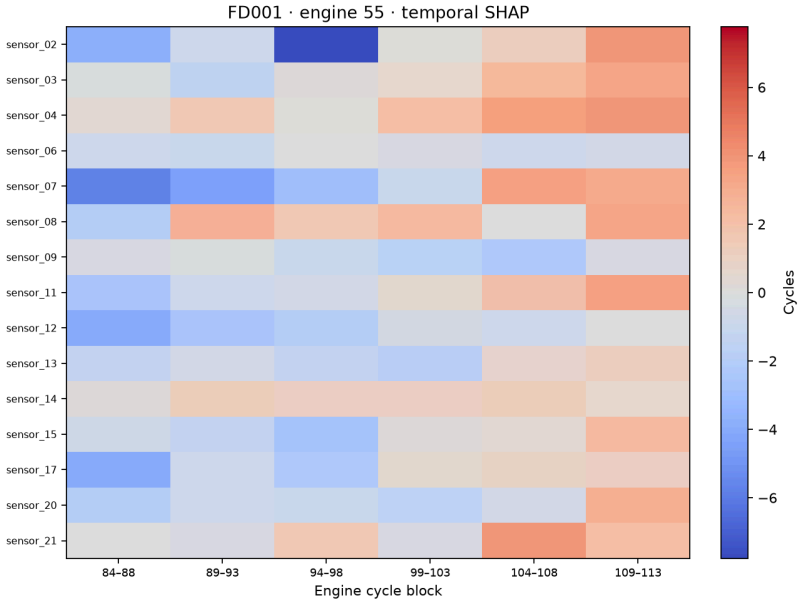


Figure 2. Temporal SHAP for FD001 engine 55, where 87.27 cycles were predicted against 137 actual cycles. FD002 engine 86 was underestimated by 97.80 cycles, while FD003 engine 16 was overestimated by 65.55.

8. Deployment, conclusion and references

The FD001 seed-42 student was exported to FP32 ONNX opset 17. CPU checks covered 100 official test windows plus a padding case, with a maximum PyTorch difference of 0.000046 cycles and raw RMSE of 17.55. That Windows CPU run recorded 0.720 ms median and 1.226 ms p95 latency. Physical Jetson latency, memory, power and energy remain pending.

The compact student improved on fixed LIF regression, while conventional models led the original RMSE comparison. The extensions show an accuracy benefit from ensembling, a capacity trade-off, and greater student sensitivity under worst tested sensor outages. Conclusions are limited to simulated data and the stated seed coverage.

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Source code, complete metric tables, executed notebooks and research evidence